Siddharth Sondhi

Email: sondhisiddharth@gmail.com
Phone: (510)-598-6588
LinkedIn: www.linkedin.com/in/siddharth-sondhi
GitHub: github.com/SiddharthSondhi
Portfolio: siddharthsondhi.github.io

Education

- **San Jose State University** — San Jose, CA
  Master of Science in Computer Science; GPA: 3.81 — *Aug. 2024 – May 2026*
- **UC Riverside** — Riverside, CA
  Bachelor of Science in Computer Science; GPA: 3.65 — *Sep. 2020 – Jun. 2024*

Projects

- **OpenGL Graphics Pipeline Renderer**: Real time renderer built in C++ using OpenGL and GLSL shaders. Implemented a full graphics pipeline featuring Blinn-Phong lighting, directional shadow mapping with PCF for soft shadows, model loading via Assimp, instanced rendering, cubemap skyboxes, framebuffer based post processing effects, and normal mapping. Designed a custom scene management system as well, supporting rendering of multiple scene by selection through GUI by integrating Dear ImGui.
- **N-Body Galaxy Simulation with Octree**: Gravitational particle simulation in C++ using Euler integration. Optimized from O(n^2) brute force to O(n log n) via a Barnes-Hut octree algorithm, enabling real time simulation of many particles. Visualized with OpenFrameworks.
- **Ray Tracer**: CPU ray tracer with reflections, refractions, and shadows. Implements Phong shading and multi bounce recursive rays to produce realistic images. Built in C++.
- **Ray Marcher**: Rendered 3D shapes defined by signed distance functions using a ray marching algorithm. Implemented in C++ and OpenFrameworks.
- **AI-Assisted VR Anatomy Education App**: Unity-based virtual reality environment integrating local LLM inference, speech recognition, and function calling pipelines for natural language interaction. Conducted a formal comparative user study between AI-assisted and traditional VR interaction methods. Developed in C# with Unity as part of MSCS thesis work.

Experience

- **Verve Group**
  Software Engineering Intern — *Summer 2023*
  - **Role**: Contributed to a data science project extracting and analyzing device and location data to derive user insights for targeted advertising. Investigated correlations between demographic data (e.g., US Census income brackets) and click behavior.
  - **Languages & Tools**: Python, Pandas, Git, JupyterLab
- **InMobi**
  Software Engineering Intern — *Summer 2021*
  - **Role**: Automated a previously manual data upload process by building a new tab in a web application. Implemented file parsing, error checking, and user feedback in JavaScript and HTML. Used Python to make API calls and upload parsed data to the company database.
  - **Languages & Tools**: JavaScript, Python/Django, Git, HTML/CSS, Docker

Skills

- **Languages**: C++, GLSL, Python, C#, Java
- **Graphics & Rendering**: OpenGL, GLSL Shaders, Real-Time Rendering, Rasterization, Graphics Pipeline, Shader Programming, Phong Lighting, Ray Tracing, Ray Marching, Signed Distance Functions (SDFs), Framebuffer & Post-Processing, Cubemaps, Instanced Rendering, Geometry Shaders, Light Transport
- **Simulation & Math**: N-Body Simulation, Particle Systems, Barnes-Hut Algorithm, Octree Spatial Partitioning, Euler Integration, Linear Algebra, 3D Math, Coordinate Transforms, Matrix Operations
- **Tools & Frameworks**: Git/GitHub, OpenFrameworks, Assimp, Dear ImGui, Unity, CMake, Command Line